

Faculty of Science
BCA II-Year, CBCS-IV Semester Regular Examinations April/May-2025

PAPER: DISTRIBUTED AND CLOUD COMPUTING

Time: 3 Hours

Max Marks: 70

Section-A

I. Answer any FIVE of the following questions

(5x4=20 Marks)

1. What is Multicast Communication? Explain. ✓
2. Write a short note on Message Queues. ✓
3. Mention the Roots of Cloud Computing. ✓
4. What do you mean by Virtual Resource Management? ✓
5. Specify some of the Public Cloud Platforms. ✓
6. How to share Resources in Distributed Systems? ✓
7. What are Request Reply Protocols? ✓
8. Write the Features of VMware. ✓

Section-B

II. Answer the following questions

(5x10=50 Marks)

9. A) Define Distributed Systems. Discuss the Trends in Distributed Systems. ✓

OR

- B) Explain Inter process Communication System Model.

10. A) Describe about Remote Procedure Calls. ✓

OR

- B) Explain the Shared Memory Approaches in RMI.

11. A) What is Cloud? Write the Types of Clouds. Describe the desired features of a Cloud.

OR

- B) Explain any Cloud Computing Service Models. ✓

12. A) What is Virtualization? Describe the Levels of Virtualization.

OR

- B) Explain Virtualization Data Centre Automation with an Example. ✓

13. A) Describe the AWS Architecture? ✓

OR

- B) Discuss on Inter-Cloud Resource Management.

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Code: 4017/R

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PAPER: NETWORK SECURITY

Time: 3 Hours

Max Marks: 70

Section-A

I. Answer any FIVE of the following questions

(5x4=20 Marks)

1. What is Cryptography?
2. What is Plaintext, Cipher Text, Public Key and Private Key?
3. What is Bit Commitment, Fair Coin Flips?
4. What is Information Theory?
5. What is Hashing?
6. What is Digital Signature?
7. Define public key management.
8. What is Digital Cash?

Section-B

II. Answer the following questions

(5x10=50 Marks)

9. a). Explain about Cryptography methods with examples.
OR
b). Explain about Cryptographic Protection of Databases.
10. a). Explain about advanced protocols with examples.
OR
b). Explain about subliminal channel, zero knowledge proof with examples.
11. a). Explain about Encrypting data for storage hardware and software Encryption.
OR
b). Explain about Electronic code book, Block replay and cipher block chaining.
12. a). Explain about Differential and Linear Crypto analysis.
OR
b). Explain about DES.
13. a). Explain about MD5 Algorithm.
OR
b). Explain about ISO authentication framework.

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Code: 4023/R

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PAPER: SOFTWARE ENGINEERING

Time: 3 Hours

Max Marks: 70

Section-A

I. Answer any FIVE of the following questions

(5x4=20 Marks)

1. Explain about Program versus Software with example?
2. What is Stakeholder Analysis?
3. Explain about Software Architecture?
4. What is Sequence Diagram?
5. What is Software Testing?
6. What is Verification versus Validation?
7. What is Cloud Computing?
8. What is Good Coding Standards?

Section-B

II. Answer the following questions

(5x10=50 Marks)

9. a) Explain about Software Development Process and its stages.
(OR)
b) Explain about Requirement Engineering Process.
10. a) Explain about IEEE Standard of Software Requirement Specifications.
(OR)
b) Explain about Validation Techniques and Requirement Management Planning.
11. a) Explain about Architectural Style and Cloud Computing.
(OR)
b) What is Data Modeling? Explain about Top Down structure and Object oriented Analysis.
12. a) Explain about Object Oriented Design using UML Diagrams.
(OR)
b) What is Decision Table? Explain about importance of Decision Table.
13. a) What is Software Testing? Explain about its Types.
(OR)
b) Explain about Software Quality? Explain about different Quality Management Models.

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Code: 4026/R

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PAPER: DATA SCIENCE USING PYTHON

Time: 3 Hours

Max Marks: 70

Section-A

I. Answer any FIVE of the following questions

(5x4=20 Marks)

1. What are the Applications of Data Science?
2. What is Lines & Indentation, Quotations in Python?
3. What is String Slice?
4. What is Numpy array?
5. Define Aggresion.
6. What is a Data Analysis Process?
7. Define void function.
8. What is List (Vs) Tuple?

Section-B

II. Answer the following questions

(5x10=50 Marks)

9. a). Explain about techniques and Methods in Data Analysis Process with Examples.
(OR)
b). Explain about Python Modes of Interpreter and Identifiers.
- 10 a). Explain about if-elif-else statement in python with example.
(OR)
b). What is Function? Explain about Types of Functions.
11. a). Explain about String and its operations in Python.
(OR)
b). What is Tuple? Explain about Tuple Operations with python Code.
12. a). Define Numpy? Explain about Numpy operations with Examples.
(OR)
b). Explain about sorting arrays with example.
13. a). What is Pandas? Explain about Pandas Operations in Python.
(OR)
b). Explain about aggregation and grouping with Examples.

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PAPER: ARTIFICIAL INTELLIGENCE

Time: 3 Hours

Max Marks: 70

Section-A

I. Answer any FIVE of the following questions

(5x4=20 Marks)

1. What is Artificial Intelligence and explain its features?
2. What is the importance of two bounds in Alpha- Beta Cutoffs?
3. What are Bayesian Networks?
4. What do you mean by Constraint Satisfaction?
5. What do you mean by Navigation? Explain.
6. Specify the Characteristics of a Problem.
7. How to represent simple facts in Logic?
8. How to Augment a Problem Solver?

Section-B

II. Answer the following questions

(5x10=50 Marks)

9. A) How to define a problem as State Space Search? Explain in detail with suitable example.
OR
B) What are Heuristics Search Techniques? Explain Hill Climbing Technique.
10. A) Explain Min-Max Search Procedure.
OR
B) Justify the need for Computable Functions and Predicates in Logic.
11. A) What is Non Monotonic Reasoning and explain.
OR
B) Describe Dempster-Shafer Theory.
12. A) What is Learning and explain
OR
B) What are Expert Systems and explain.
13. A) Describe Speech Recognition process in detail.
OR
B) What is Natural Language Processing and explain.
